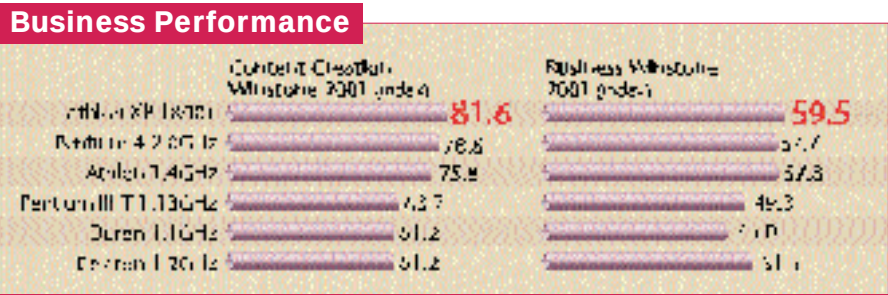


few seconds more to perform a particular task, it still amounts to fairly large idle time if you add up all the extra seconds your PC has taken to perform different tasks. For instance, if a processor takes only 3 or 4 seconds to run, say, Microsoft Word, it would seem to be no big deal. But if you add up all the extra seconds it takes to perform tasks, it will amount to some serious time you'd spend waiting while doing the paste job from Excel to Word, or while zipping up large documents.

In the Elite segment we found that in most cases the Athlon XP series performs far better than the Pentium 4. This clearly shows that despite the fact that the Athlon XP processors are slower 'clock for clock' than the Pentium 4 series, in real-world terms they perform more work in the same number of CPU cycles. They score better.



In some cases the Athlon (Thunderbird) even beat the Pentium 4 processor and gave some serious competition to big brother, the Athlon XP. The Thunderbird-based Athlon 1.4 GHz scored 75.8 in Content Creation Winstone 2001, while the Athlon XP 1600+ which also runs at 1.4 GHz internally, returned a score of 78.4. This clearly indicates that the Palamino core-based Athlon XP has been tweaked to an extent. It is not just a simple switch

to the new naming convention without a real gain in performance.

Though the Athlon XP won by a fair margin, it can be argued that once you cross 70 in Content Creation Winstone 2001, there is little discernible difference from one processor to the other. Nevertheless, the top-of-the-line Athlon XP 1800+ zipped past all the others with a huge score of 81.6. This is a record of sorts and no other processor we tested came even close to this magical figure. In the Business Winstone 2001 test, we discovered a fairly close mix of results with all the top contenders scoring within a couple of points of each other.

In the performance segment, the Pentium III 1.13 GHz (Tualatin) scored a decent 63.7 in Content Creation Winstone 2001, whereas the other Pentium IIIs scored in the range of the 40s. Even here, the Athlon family continued to

impress with the Athlon 1.13 GHz scoring a hefty 66.6. The results were more interesting in the Business Winstone 2001 test as only the Athlon family could maintain scores over 50. The Pentium III 1.13 GHz finished at 49.3.

In the Value segment, the GigaHertz variants of the Celeron and the Duron showed respectable scores of 61.2 each in Content Creation Winstone 2001. The Celeron 1.2 GHz proved to be the saving grace for the Celeron series as the Celeron 800 scored an extremely low 33.9 in the Business Winstone 2001 test (the lowest in this test). Celeron 1.2 GHz pummelled the Duron 1.1 GHz by scoring 51.5 as against the 45 in the Business Winstone 2001 test.

One thing visible in these tests was that the GigaHertz variants of all the processors scored respectably well—if you need to run only office applications and tasks, any of these processors would fit the bill.

The Pentium III 700 and the Celeron 800 suffered in both of these tests primarily because of their low clock speeds. It

Glossary

32-bit processors: 32-bit represents the width of the registers within the CPU core. A register is rather like a holding area for data. All the processors reviewed in this comparison test are 32-bit.

Bus: A set of lines used to move information from one part of the computer to another. Typically, a bus is divided into three functional groups: data, address, and control.

Clock cycle: The basic unit of processor activity. This refers to the speed at which an integrated chip processes data. Everything needs to be sequenced in computers and the clock speed determines how fast all of this will be done.

FPU: Floating Point Unit handles calculations that use or produce non-integer numbers. Instructions to be processed by the FPU are known as the x87 set.

FSB: Front Side Bus is the external frequency provided to the processor.

Instructions set: The set of rules that tells the CPU what to do with each piece of data it gets. All processors use the basic x86 set as a starting block, but each successively higher generation chip utilises additional instructions. All processors these days have extensions to the fundamental rules that are used for complex things such as video, audio and mathematics. Intel calls theirs MMX and SSE/2, whereas AMD have labeled theirs as 3DNow!

Multiplier: This number decides the core internal frequency of the processor. For example, a multiplier of 10 with the FSB at 133 MHz will yield an internal core frequency of 1330 MHz for the processor.

Transistor: A transistor is an electronic switch. The speed at which a processor can operate at is determined by how quickly the transistors can switch between their two states, which can be increased by reducing their size and putting more of them in the processor.

To make smaller circuit sizes, the transistors need to be made smaller too. When we hear 0.18 micron fabrication technology it primarily means that transistor gates etched on the silicon have a dimension of only 0.18 microns. By reducing this size one can physically etch more transistors into the same piece of silicon.

Vcore: This is the operating voltage of the CPU core.

The launch of the Pentium II/III and the K6 series from AMD. The competition heats up.

Sixth Generation 2000

1997 AMD introduces the Athlon (K7), follows it up with Thunderbird which successfully competes against the P-III. The battle now moves to Intel's (P7/Willamette) Pentium 4 and the Athlon XP (Palamino).